

SUPPORTABILITY ENGINEERING

Volume 5

Compliance by Design:
Supportability Engineering
in Regulated Environments

How the SE framework satisfies SOC 2, ISO 27001, ISO 42001, GDPR, SOX, and FedRAMP — and turns compliance evidence into operational readiness

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The Audit Finding Nobody Expected

The engineering team had done everything right. The system had been running in production for eight months without a major incident. The code was clean. The tests passed. The deployment pipeline was automated.

Then the auditors arrived.

They asked a simple question: for each system that processes personal data, can you show us the documented failure modes, the evidence that those failure modes were tested before go-live, and the escalation procedure if a data breach occurs?

The engineering team looked at each other. The failure modes were in someone's head. The pre-go-live testing had happened, but nobody had documented what was tested or what the results were. The escalation procedure existed in a Confluence page that hadn't been updated in six months and didn't mention the GDPR 72-hour notification requirement.

The auditors issued a finding. Not because the system was insecure. Not because a breach had occurred. Because there was no evidence that the organisation had systematically thought through what failure looked like and what they would do about it.

“We passed every technical control. We failed on process documentation. The auditors weren't asking whether our system was safe. They were asking whether we could prove it was safe — and we couldn't.”

That organisation spent the next three months producing documentation retrospectively. It cost more than building it forward would have — and it was less accurate, because it described a system that had already changed.

That is the problem Supportability Engineering solves. And in regulated environments, it solves two problems simultaneously: it makes systems operable, and it produces the audit evidence that proves they were built with control in mind.

Executive Summary

Compliance frameworks and supportability frameworks appear to operate in different worlds. Compliance is the language of auditors, control objectives, and evidence. Supportability is the language of engineers, incident response, and operational readiness. In most organisations they are managed by different teams, governed by different processes, and reported to different executives.

They are, however, solving the same underlying problem: ensuring that when something goes wrong, the right people know about it, can respond to it, and can demonstrate that it was handled correctly.

Supportability Engineering — a six-phase shift-left framework for designing observability, failure handling, and operational readiness into systems before they are built — produces deliverables that directly satisfy requirements across SOC 2, ISO 27001, ISO 42001, GDPR, SOX, and FedRAMP. Not as a side effect. As a primary output.

This volume explains how. It is written for two audiences simultaneously:

Engineering and DevOps teams	You are building systems in regulated environments and need to understand how SE phase deliverables map to the compliance requirements your organisation is subject to. This volume gives you that mapping and shows how doing SE well means doing compliance well — without running two separate processes.
Compliance and audit teams	You are responsible for ensuring controls exist and can be evidenced. This volume shows you how SE deliverables — signed SRDs, SARs, SICs, STPs, SRRs, and SFLs — function as control documentation, change management evidence, and audit trail artefacts. A team running SE is a team that is producing the evidence you need.

The volume is structured in three parts. Part one establishes why compliance and supportability are the same problem expressed in different vocabularies. Part two maps each of the five compliance frameworks to the SE phases that satisfy their requirements. Part three describes the C- prefix compliance extension layer — a set of additions to each SE deliverable that completes the compliance picture without duplicating work.

Section 1: Compliance and Supportability Are the Same Problem

Every compliance framework that governs software systems is ultimately asking four questions. They use different language, different control numbers, and different audit procedures — but the underlying questions are the same.

The Compliance Question	Compliance Language	SE Answer
What can go wrong?	Risk and threat modelling	SE: Failure mode inventory (SRD §1.4), dependency risk assessment (SAR §2.7)
Will you know when it does?	Monitoring, alerting, and detection controls	SE: Observability requirements (SRD §1.3), four golden signals (SIC §3.2), alert validation (STP §4.4)
Can you respond correctly?	Incident response procedures and escalation	SE: Escalation path definition (SRD §1.8), runbook walkthrough (STP §4.6), on-call simulation (STP §4.7)
Can you prove it?	Audit evidence, change records, signed approvals	SE: Signed deliverables at every phase — SRD, SAR, SIC, STP, SRR, SFL — each version-controlled

The reason most engineering teams experience compliance as a burden is that they answer these questions twice — once operationally, through their engineering practices, and once for the auditors, through documentation produced retrospectively. Supportability Engineering produces the answer once, at the right time, in a form that serves both purposes.

Where SE Deliverables Sit in the Control Framework

Compliance frameworks organise controls into categories. The mapping below shows where SE deliverables land in the most common control categories — before we get to framework-specific requirements.

Control Category	SE Deliverable Coverage
Change Management	SRR sign-off (Support Lead + Engineering Lead) is a formal change approval record. Version field on every deliverable creates a change history. SRD currency confirmation before each phase is a change gate.
Risk Assessment	SRD failure mode inventory and business impact pre-classification constitute a documented risk assessment for every feature. SAR failure point map and observability gap list extend this to architectural risk.
Access and Data Controls	SIC §3.2 requires explicit verification that no sensitive data appears in logs. SRD §1.8 flags regulatory and compliance obligations including PII, financial data, health data, and audit trail requirements.
Incident Response	STP on-call simulation is a documented rehearsal of the incident

	response procedure. SRR communication templates are pre-approved incident communications. SFL incident scoring is a post-incident record.
Audit Trail	Every SE deliverable carries Name / Signature / Version / Date. The chain from SRD through SFL is a timestamped, version-controlled record of every decision made about a feature from requirements to production.
Vendor / Third-Party Risk	SAR dependency risk assessment documents every external dependency, its failure modes, and detection capability. This is the evidence base for third-party risk management requirements.

Section 2: Five Frameworks — What Each Requires, What SE Provides

SOC 2

SOC 2 is organised around the Trust Services Criteria. The criteria most directly relevant to software systems in production are availability, confidentiality, and security — with supporting criteria around change management, risk assessment, and incident response.

SOC 2 Criterion	SE Coverage
CC7.1 — Detection of security events	SIC golden signal instrumentation and STP alert validation demonstrate that detection capability was designed in and verified before go-live.
CC7.2 — Monitoring for anomalies	SRD observability requirements and SAR observability gap list constitute the documented monitoring design. STP dashboard verification confirms it works.
CC7.3 — Incident response procedures	SRD escalation path definition plus STP on-call simulation satisfy the requirement for documented and rehearsed incident response procedures.
CC8.1 — Change management	SRR dual sign-off is a formal change approval. Version-controlled deliverables from SRD through SFL constitute the change record. SRD currency gate means no design proceeds without current requirements.
A1.2 — Availability commitments	SRD business impact pre-classification maps customer segments to SLA commitments. This is the documented basis for availability monitoring thresholds.
C1.1 — Confidentiality controls	SIC §3.2 sensitive data check and SRD §1.8 compliance flags together constitute evidence that confidentiality was considered at requirements and verified at build.

ISO 27001

ISO 27001 requires a documented Information Security Management System (ISMS) with evidence of risk assessment, treatment, and operational controls. The Annex A controls most relevant to software systems span asset management, operations security, incident management, and supplier relationships.

ISO 27001 Control	SE Coverage
A.12.1.2 — Change management	SRR sign-off and the phase-gate chain (SRD → SRR) satisfy the requirement for formal change approval and documented change records.
A.12.4 — Logging and monitoring	SRD observability requirements, SIC logging implementation checklist, and STP log quality review together constitute a documented logging

	and monitoring control.
A.16.1 — Incident management procedures	SRD escalation path, STP on-call simulation, SRR communication templates, and SFL incident scoring collectively satisfy the incident management procedure requirement.
A.14.2.8 — System security testing	STP failure injection testing is documented evidence of pre-release security and resilience testing. The STP pass/fail recommendation is the test result record.
A.15.1 — Supplier relationships	SAR dependency risk assessment documents third-party dependencies, failure modes, and detection capability — the evidence base for supplier relationship controls.
A.18.1 — Compliance with legal requirements	SRD §1.8 regulatory flags plus the C-SRD compliance obligations section (see Section 4) constitute the documented compliance requirement identification for each feature.

ISO 42001

ISO/IEC 42001 is the first certifiable AI Management System (AIMS) standard. It follows the same Harmonized Structure as ISO 27001, clauses 4 through 10 covering context, leadership, planning, support, operation, evaluation, and improvement, plus Annex A, a set of AI-specific controls spanning AI policy, internal organisation, resources, impact assessment, AI system lifecycle, data, information for interested parties, responsible use, and third-party relationships.

Where ISO 27001 asks whether information is secure, ISO 42001 asks whether an AI system is governed responsibly across its lifecycle, including dimensions SE does not natively score: bias, fairness, and societal impact. SE's contribution is the operational layer underneath the certification layer, the mechanics that make an AI system diagnosable and supportable in production, which ISO 42001 assumes exist but does not specify in build-level detail.

ISO 42001 Domain	SE Coverage
Annex A.5 — AI system life cycle	The SRD through SRR phase chain maps directly to AI system planning, design, verification, deployment, and operation. Each lifecycle stage already carries a signed SE deliverable.
Annex A.6 — Data for AI systems	SRD observability requirements and SAR trace boundary definition extend naturally to data lineage and quality tracking for training, validation, and production data feeding an AI feature.
Annex A.7 — Information for interested parties	SRD feature description and failure mode inventory, combined with SRR communication templates, satisfy the requirement to disseminate purpose, limitations, and monitoring capability to users and stakeholders.
Annex A.8 — Use of AI systems	SIC failure mode unit tests and STP failure injection testing, extended to AI-specific failure modes such as low-confidence or unverified output, provide the human oversight and operational control evidence this objective requires.
Annex A.9 / A.10 — Third-	SAR dependency risk assessment extends to AI model and data

party and supplier relationships	suppliers, covering the same due-diligence ground Annex A.10 requires for AI-specific vendors.
Clause 9 — Performance evaluation	SFL incident scoring and the quarterly supportability review provide the monitoring, measurement, and management review evidence ISO 42001 requires on an ongoing basis.

ISO 42001 also requires governance elements SE does not provide on its own: a named AIMS owner, a Statement of Applicability, and a formal AI policy. These are organisational governance artefacts, not engineering deliverables, and they sit alongside SE rather than inside it. The C-ISO42001 extension below closes the gap between the two.

GDPR

GDPR imposes specific obligations on organisations that process personal data — obligations that directly affect how software systems are designed, monitored, and operated. The most operationally significant are data protection by design, breach detection and notification, and the ability to demonstrate compliance.

GDPR Obligation	SE Coverage
Article 25 — Data protection by design	The SRD is the mechanism for data protection by design. When SRD §1.8 flags personal data involvement, observability requirements must include data access logging and the failure mode inventory must include data exposure scenarios. This is design-time, not retrofit.
Article 32 — Security of processing	SIC §3.2 (no sensitive data in logs) and SRD §1.3 (observability requirements including data access events) together constitute evidence of technical measures to ensure appropriate security.
Article 33 — Breach notification (72 hours)	SRD escalation path must define the data protection officer notification trigger and timeline. SRR communication templates must include a GDPR breach notification template. STP on-call simulation must rehearse the 72-hour notification path explicitly.
Article 35 — Data Protection Impact Assessment	The SRD failure mode inventory, business impact pre-classification, and compliance flags section together constitute the SE-native equivalent of a DPIA for features involving personal data. The C-SRD extension formalises this mapping.
Article 5(2) — Accountability principle	The full chain of signed, version-controlled SE deliverables from SRD to SFL is the accountability documentation GDPR requires — evidence that data protection obligations were considered, addressed, and verified at each phase.

SOX

Sarbanes-Oxley applies to publicly traded companies and imposes requirements on the integrity of financial reporting systems and the IT controls that support them. Section 404 — the requirement for

management assessment of internal controls over financial reporting — is the primary driver of SOX IT compliance work.

SOX Requirement	SE Coverage
IT General Controls — Change management	Every feature that touches financial reporting systems requires a documented change approval chain. SRR dual sign-off, combined with the version-controlled phase deliverables, constitutes that chain. A feature that shipped without a signed SRR is a change management control failure under SOX.
IT General Controls — Access controls	SRD §1.8 flags features involving financial data. SIC explicitly checks for sensitive data in logs. The combination is evidence that access to financial data was considered and controlled at design and build.
IT General Controls — Operations	SRD escalation path definition and STP runbook walkthrough satisfy the operational procedure documentation requirement. The SFL incident scoring log satisfies the requirement for documented evidence of how incidents affecting financial systems were handled.
Section 302 — Management certification	Executives certifying financial statement accuracy need assurance that the systems supporting those statements have documented, tested controls. SE deliverables are that assurance — a complete, auditable record from requirements to production.

FedRAMP

FedRAMP is the US federal government's authorisation framework for cloud services. It is based on NIST SP 800-53 and imposes some of the most rigorous documentation and evidence requirements of any compliance framework. FedRAMP authorisation requires a System Security Plan, continuous monitoring, and documented incident response.

FedRAMP Control (NIST 800-53)	SE Coverage
SI-2 / SI-3 — Flaw remediation and malicious code protection	SIC failure mode unit tests and STP failure injection testing constitute documented evidence of pre-release flaw identification and remediation testing.
AU-2 / AU-12 — Audit events and audit generation	SRD observability requirements and SIC logging checklist define and implement the audit event set. STP log quality review verifies it. This maps directly to the FedRAMP requirement to define, implement, and verify audit logging.
IR-3 / IR-4 — Incident response testing and handling	STP on-call simulation is the documented incident response test FedRAMP requires annually. SFL incident scoring records are the evidence of ongoing incident handling capability.
CM-3 / CM-4 — Configuration change control and impact analysis	SRR is the configuration change control gate. SAR failure point map and observability gap list constitute the impact analysis for architectural changes. Version-controlled deliverables are the change record.
CA-7 — Continuous	SFL observability gap log and quarterly supportability review satisfy the

monitoring	continuous monitoring documentation requirement — evidence that monitoring is ongoing and gaps are tracked to resolution.
SA-11 — Developer security testing	STP is the developer security testing documentation FedRAMP requires — failure injection, log quality review, alert validation, runbook walkthrough, all with signed pass/fail results.

Section 3: The SE Phase Overlay for Regulated Environments

The five framework mappings in Section 2 show what SE already covers. This section defines what regulated environments need in addition — the specific additions to each SE phase that complete the compliance picture. These additions are the C- prefix extension layer described in Section 4.

The guiding principle is the same principle that governs the base SE framework: ask the compliance question at the phase where it is cheapest to answer. A data residency requirement identified in the SRD is a design constraint. The same requirement identified after deployment is a remediation project.

Phase 1 — C-SRD: Compliance Requirements Document Extension

The base SRD flags compliance obligations in §1.8. The C-SRD extension expands this into a full compliance requirements section that sits alongside the base SRD.

WHAT THE C-SRD ADDS

Applicable framework identification: Which frameworks apply to this feature — SOC 2, ISO 27001, GDPR, SOX, FedRAMP — and the specific control references that are triggered by this feature's data handling, user access, and financial system involvement.

Data classification: Formal classification of all data this feature processes, stores, or transmits — not just a yes/no flag. Category, sensitivity level, residency constraints, and retention requirements.

DPIA trigger assessment: For GDPR-regulated features: explicit determination of whether a Data Protection Impact Assessment is required and, if so, confirmation that one has been initiated.

Compliance sign-off: A named compliance officer or DPO confirms the compliance requirements are complete and correctly identified before design proceeds. This is the compliance equivalent of the support lead gate.

Phase 2 — C-SAR: Compliance Architecture Review Extension

The SAR reviews architecture for supportability. The C-SAR extends this review to include compliance-specific architectural requirements.

WHAT THE C-SAR ADDS

Data flow mapping: Where personal or regulated data flows within the architecture — entry points, storage locations, processing components, exit points. Required for GDPR Article 30 records of processing activities.

Encryption and data protection controls: Which components handle regulated data, what encryption is applied in transit and at rest, and whether the architecture creates any unencrypted exposure points.

Compliance blind spots: Architectural points where compliance-relevant events — data access,

privilege escalation, configuration change — are not observable. These are added to the observability gap list with compliance priority.

Audit trail integrity: Whether the audit log chain is tamper-evident, complete, and independently verifiable — required for SOX, FedRAMP, and ISO 27001.

Phase 3 — C-SIC: Compliance Implementation Checklist Extension

The SIC verifies supportability standards at build. The C-SIC adds a compliance verification gate to the PR review process.

WHAT THE C-SIC ADDS

Regulated data in logs — zero tolerance: The base SIC checks for sensitive data in logs. The C-SIC makes this explicit for each regulated data category identified in the C-SRD. The reviewer signs off on each category individually.

Audit event completeness: Every compliance-relevant event identified in the C-SRD is instrumented and verified at the code level. Not just logging standards — specific events required by the applicable frameworks.

Data retention and deletion: If the feature stores regulated data, the retention policy is implemented and the deletion mechanism is tested. GDPR right-to-erasure compliance begins here.

Compliance reviewer sign-off: For features touching regulated data, a compliance-aware reviewer explicitly signs the C-SIC in addition to the standard code reviewer.

Phase 4 — C-STP: Compliance Test Plan Extension

The STP validates operational readiness. The C-STP extends this to validate compliance control effectiveness.

WHAT THE C-STP ADDS

Compliance scenario testing: Deliberately trigger each compliance-relevant failure mode: data exposure attempt, privilege escalation attempt, configuration change without approval, regulatory notification trigger. Verify detection, logging, and response.

72-hour notification rehearsal (GDPR): For GDPR-regulated features, the on-call simulation explicitly rehearses the breach detection-to-notification path. The simulation timer is set to 72 hours. The DPO notification step is verified.

Audit log verification: Confirm that audit logs are complete, tamper-evident, and correctly formatted for the applicable frameworks. Verify that audit events are retained for the required period.

Evidence package: The C-STP produces a signed evidence package — test scenarios, results, and reviewer names — that functions directly as audit evidence for pre-release security testing requirements.

Phase 5 — C-SRR: Compliance Readiness Review Extension

The SRR is the final gate before production. The C-SRR adds a compliance readiness confirmation to the dual sign-off.

WHAT THE C-SRR ADDS

Compliance officer sign-off: A third signature on the SRR from the compliance officer or DPO. Three parties now confirm go-live readiness: Support Lead, Engineering Lead, and Compliance. This is the change approval record for regulated environments.

Regulatory notification readiness: Confirm that notification procedures are in place for each applicable framework — GDPR 72-hour path, SOX material weakness reporting, FedRAMP incident reporting to US-CERT.

Data processing records current: Article 30 records of processing activities updated to include this feature before go-live. Not after. This is the GDPR accountability principle in practice.

Audit evidence package complete: All C- prefix deliverables from C-SRD through C-STP are signed, version-controlled, and stored in the designated audit evidence repository before the feature ships.

Phase 6 — C-SFL: Compliance Feedback Loop Extension

The SFL converts operational experience into framework improvements. The C-SFL adds a compliance dimension to incident scoring and the quarterly review.

WHAT THE C-SFL ADDS

Compliance incident classification: Every incident is assessed for compliance implications at the time of scoring — not retrospectively. Was regulated data potentially exposed? Was a notification obligation triggered? Was a compliance control bypassed?

Regulatory notification log: A dedicated record of every notification made to regulators or affected data subjects. Required for GDPR, FedRAMP, and SOX. Maintained in the SFL and reviewed quarterly.

Control effectiveness scoring: Alongside the standard supportability score, each incident is scored on whether the compliance controls performed as designed. Low scores feed directly into the C-SRD for the next cycle.

Quarterly compliance review: The standard SFL quarterly review adds a compliance agenda item: open findings, control gaps, regulatory notifications made in the period, and any changes to the applicable framework requirements.

Section 4: SE Deliverables as Audit Evidence

A recurring challenge in compliance programmes is the gap between operational practice and audit evidence. Teams do the right things — they test their systems, they review their code, they run incident drills. Then the auditors arrive and ask for documentation, and the documentation either doesn't exist or doesn't match what actually happened.

SE eliminates this gap by making the documentation a byproduct of the work, not a separate activity performed for the auditors.

The Audit Evidence Chain

A fully executed SE programme — SRD through SFL, including C- prefix extensions — produces the following audit evidence automatically:

Audit Evidence Required	SE Deliverable That Provides It
Risk assessment record	SRD failure mode inventory + business impact pre-classification + C-SRD compliance requirements. Timestamped, version-controlled, signed by product, engineering, support, and compliance.
Change management record	The phase gate chain — SRD currency confirmed before SAR, SAR open items resolved before SIC, STP pass before SRR — is a documented, signed change approval chain. SRR is the change authorisation record.
Security testing evidence	STP failure injection results, log quality review, alert validation, C-STP compliance scenario testing, and signed pass/fail recommendation. This is pre-release security testing documentation.
Incident response procedures	SRD escalation path + STP on-call simulation + SRR communication templates = documented, rehearsed, and pre-approved incident response. The simulation record is the test evidence.
Operational monitoring controls	SRD observability requirements + SIC golden signal implementation + STP dashboard verification = documented monitoring control design, implementation, and verification.
Post-incident records	SFL incident scores, observability gap log, runbook accuracy tracking, regulatory notification log (C-SFL), and quarterly review records = ongoing operational control evidence.

What Auditors Actually Look For

Experienced compliance auditors are not looking for perfect systems. They are looking for evidence of systematic thinking — that the organisation identified risks, designed controls, tested those controls, and monitors their effectiveness over time. The SE framework produces exactly this evidence at every phase.

“The question auditors ask is not whether something went wrong. It is whether the organisation had a reasonable process for preventing and detecting things going wrong. SE is that process, documented.”

The C- prefix extensions add the compliance-specific vocabulary that bridges SE language to audit language — mapping failure modes to control objectives, escalation paths to notification procedures, and incident scores to control effectiveness evidence.

Storage and Retrieval

For SE deliverables to function as audit evidence, they must be stored in a way that satisfies the tamper-evidence and retention requirements of the applicable frameworks. The following minimum standards apply:

Requirement	SE Approach
Version control	Every deliverable must carry Name / Signature / Version / Date. Version history must be maintained. The current version of each deliverable must be distinguishable from prior versions.
Access control	Audit evidence must be protected from unauthorised modification. SE deliverables should be stored in a system with access logging — a document management system, a compliance platform, or a version-controlled repository with commit history.
Retention period	SOC 2: 12 months minimum. ISO 27001: defined by the ISMS. GDPR: duration of data processing plus applicable statute of limitations. SOX: 7 years for financial system records. FedRAMP: 3 years post-authorisation.
Retrieval on demand	Auditors require the ability to retrieve a specific deliverable for a specific feature at a specific point in time. The naming convention — feature name, deliverable type, version, date — must support this retrieval.

Section 5: Implementation — Running SE in a Regulated Environment

Running SE in a regulated environment does not require a separate compliance programme running in parallel. It requires three things: identifying which C- prefix extensions apply to each feature, integrating the compliance sign-off into the existing gate structure, and storing deliverables in a way that satisfies the retention and retrieval requirements of the applicable frameworks.

Step 1: Compliance Scope at the SRD

The first question in every SRD session for regulated environments is: which compliance frameworks apply to this feature? The answer determines which C- prefix extensions are activated. A feature that handles no personal data, has no financial system involvement, and is not part of a FedRAMP boundary needs only the base SE deliverables. A feature that processes payment card data in a FedRAMP-authorized environment activates the full C- extension stack.

Scope Profile	C- Extensions Required
No regulated data, no financial systems, no government cloud	Base SE deliverables only. No C- extensions required.
Personal data involved (any volume)	C-SRD (DPIA assessment), C-SIC (data in logs), C-SRR (Article 30 update, DPO sign-off). GDPR 72-hour path in C-STP if breach notification could be triggered.
AI system in scope	C-SRD (AI impact assessment trigger, framework identification), C-SAR (data flow mapping for training/inference data), C-SIC (compliance reviewer sign-off for AI-specific events), C-SRR (AIMS readiness confirmation). See Vol. 2 for AI-specific failure modes that feed this scope profile.
Financial reporting system involvement	C-SRD (SOX scope flag), C-SRR (SOX change approval record, three-signature sign-off). SOX material weakness notification path in C-STP.
FedRAMP authorisation boundary	Full C- extension stack. C-STP evidence package required. C-SFL regulatory notification log required. All deliverables stored per FedRAMP retention requirements.
SOC 2 or ISO 27001 audit scope	C-SRD (framework-specific control mapping), C-STP (audit evidence package), C-SRR (audit evidence repository confirmation). C-SFL quarterly review includes audit preparation agenda item.

Step 2: The Compliance Sign-Off Integration

The base SE framework has four sign-off parties at the SRD (Product Owner, Engineering Lead, Support Lead, QA Lead) and two at the SRR (Support Lead, Engineering Lead). In regulated environments, a compliance officer or DPO is added as a sign-off party at two points: the C-SRD

(confirming compliance requirements are complete before design proceeds) and the C-SRR (confirming compliance readiness before go-live).

This is not a new meeting or a new process. The compliance officer attends the existing SRD session and the existing SRR review. Their sign-off is an additional line on the existing sign-off block. The gate logic is unchanged: if any required party cannot sign, the phase does not proceed.

Step 3: Audit Evidence Repository

SE deliverables become audit evidence the moment they are signed. They need to be stored somewhere auditors can access them, in a form that demonstrates they have not been modified since signing. The minimum requirement is a document management system with version history and access logging. The recommended approach for FedRAMP environments is a version-controlled repository with cryptographic commit signing.

The naming convention for audit evidence storage: `[FeatureName]-[Deliverable]-v[Version]-[YYYY-MM-DD]`. Example: `PaymentProcessor-C-SRD-v2-2026-03-15`. This supports retrieval by feature, deliverable type, version, and date — the four axes auditors most commonly query.

What SE Does Not Replace

SE is not a complete compliance programme. It produces control evidence for the operational and development controls that directly affect software systems. It does not replace:

What SE Does Not Replace	Why
Information Security Policy	The organisational policies that govern security — acceptable use, access control policy, incident response policy — are outside SE scope. SE implements and evidences the operational application of those policies.
AI Policy and Algorithmic Impact Methodology	The AI policy ISO 42001 requires, and the methodology for assessing bias, fairness, and societal impact, are governance and ethics deliverables, not engineering ones. SE identifies where an AI impact assessment is triggered and provides the operational evidence around it. It does not author the assessment methodology itself.
Risk Register	The enterprise risk register captures strategic and operational risk. SE failure mode inventories feed into the risk register but do not replace it.
Penetration Testing	SE STP failure injection testing is not a penetration test. It validates that monitoring and response controls work. External penetration testing remains a separate requirement for most compliance frameworks.
Legal and Privacy Counsel	GDPR DPIA completion, privacy notices, data processing agreements, and regulatory correspondence require legal expertise. SE identifies the need and provides the operational evidence — it does not provide legal advice.

Section 6: Where This Volume Fits — The Five-Volume Framework

Supportability Engineering is published as a five-volume series. Each volume applies the same six-phase framework to a different operational context. Volume 5 is not a standalone compliance framework — it is the compliance extension of a complete operational readiness methodology.

Volume	Title	Scope
Vol. 1	Why the Best Support Organizations Shift Left	The foundational framework. Traditional software. The six phases, the cost curve, the case for building supportability in from the start.
Vol. 2	Shifting Left When the System Can Think	SE applied to agentic AI as the product. Six new failure categories including silent confident failure and context window drift. A- prefix deliverables. Pairs directly with Vol. 5's ISO 42001 mapping for any team shipping an AI system in scope of both frameworks.
Vol. 3	When the Builder Can't Sign Off	SE applied to AI-generated code. Four broken assumptions when AI writes the code. D- prefix deliverables.
Vol. 4	When the AI Running Your Support Needs Supporting	SE applied to AIOps and autonomous incident response tools. Seven operational categories. O- prefix deliverables. AOSR as 7th phase.
Vol. 5	Compliance by Design	SE in regulated environments. C- prefix extensions. Six framework mappings: SOC 2, ISO 27001, ISO 42001, GDPR, SOX, FedRAMP. Audit evidence chain. This volume.

The five volumes can be read independently. A team running an AIOps platform in a FedRAMP-authorized environment will use Volumes 4 and 5 together. A startup building its first AI-powered product will start with Volume 1. An enterprise engineering team working on AI-generated code in a SOC 2 environment has material across Volumes 1, 3, and 5.

The framework is designed to compose. The phase structure, deliverable naming, sign-off model, and living document framing are consistent across all five volumes. A team that learns the SE framework in one context can apply it in any other.

The Question Worth Asking

If your organisation has had a compliance finding in the last two years that traced back to inadequate documentation of a software system's failure modes, escalation procedures, or testing evidence — the question worth asking is not how to produce better documentation next time.

The question is: at what point in the development of that system was the information that would have satisfied the auditors available, and why wasn't it captured?

The answer is almost always the same. The information existed. The engineers knew the failure modes. The architects understood the data flows. The support leads had the escalation procedures. Somewhere between the design meeting and the audit, the knowledge that would have satisfied the control requirements was lost — because there was no process to carry it forward in a form that served both operational and compliance purposes.

“Compliance documentation produced retrospectively describes a system that no longer exists. Compliance documentation produced forward — as a byproduct of building the system correctly — describes the system that actually runs in production.”

Supportability Engineering is that process. In regulated environments, it is also your compliance programme for the controls that matter most: the ones that determine whether your systems can be understood, diagnosed, and reported on when something goes wrong.

That is what is on offer here. Not a compliance checklist. A methodology that makes compliance evidence a natural output of building software the right way from the first line of code.

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